ClimateAdapt Pune: Evidence-Led Climate Action Report

Pilot Study: Pune Urban Core, Maharashtra | Generated: 19 October 2026 06:42 UTC

Executive Summary: This report translates real municipal drainage statistics (OpenCity Pune), OpenStreetMap infrastructure assets (5,779 trees, 267 medical facilities, waterways), Open-Meteo 7-day atmospheric forecasts, and Google Gemini AI into actionable climate resilience priorities for Pune's electoral wards. All findings distinguish measured infrastructure facts from model proxies, with automated MILP budget optimization and expert LLM strategic synthesis.

1. High-Priority Wards (Screening Indicators)

Scores are normalized 0-100 screening proxies. Higher values denote greater exposure or infrastructure deficit.

Ward / Prabhag	Heat Proxy	Flood Proxy	Water Stress	Green Deficit	Composite
Vadgaon Sheri - Ganesh N	58.4	75.2	52.0	95.0	68.8
Arunkumar Vaidya Stadium	64.4	54.8	60.0	95.0	65.0
Ghorpade Garden	63.8	54.6	60.0	93.3	64.4
Upper Indiranagar	58.4	59.1	60.0	95.0	64.4
Mohammadwadi	58.4	60.7	52.0	95.0	63.7
Vimannagar Sanjay Park	58.4	55.4	60.0	95.0	63.1

2. Prioritized Adaptation Action Plan (Budget Allocation)

Target Portfolio Budget: INR 1,000,000 | **Committed Allocation:** INR 480,000 (48.0% efficiency)

Intervention	Category	Units	Cost (INR)	Dept	Lead Time
Cool Roof Reflective Coating	Heat & Urban Co	4 building	■480,000	PMC Building & Dev	1 mo

3. Climate Stress Scenario Comparisons

Scenario	Heat Index	Flood Index	Water Stress	Composite Index
Current Baseline	58.4	34.2	60.0	55.7
Extreme Heatwave (+3.5°C)	78.4	34.2	75.0	65.0
Extreme Monsoon Cloudburst	48.4	59.2	50.0	63.0
Post-Adaptation Plan	53.6	34.2	60.0	54.0

4. Responsible Use & Data Quality Disclosures

- ***Distinction of Evidence:*** *Drainage statistics are verified 2015-2016 OpenCity PMC releases. Atmospheric values represent Open-Meteo multi-variable NWP forecast models and must not be cited as permanent climate normals.*
- ***Explicit Missing Data:*** *Local observation wells (GSDA telemetry) and uncalibrated DEM hydrological runoff grids remain unconfirmed and are strictly excluded from automated physical simulation to prevent false certainty.*
- ***Human Review:*** *Action allocations are exploratory decision-support calculations and require municipal site surveys before civil construction.*

5. AI Ward Resilience Brief: Ward #75 (Katraj Dairy)

Analysis Engine: Google models/gemini-3.8-flash | **Zone:** Dhanakawadi | **Evaluation Date:** 2026-10-19

1. What's Life Like in This Ward?

In Katraj Dairy, everyday life gets really tough during extreme weather. In summer, the streets and houses turn into heat traps because there are barely any shade trees or parks. During monsoon, even light rain quickly fills up the roads because 60% of streets have absolutely no drains to carry the water away.

2. Problems and What We Can Do About Them

Problem	How Bad Is It?	Why Does This Happen Here?	What Can Be Done?
Streets flooding with rainwater	Very Bad	Only 40.0% of roads have drains. Rain has nowhere to go.	Clean out clogged drains and put iron trash guards so water keeps flowing.
Unbearable summer heat	Bad	No shade trees, hot tin and concrete roofs trap the sun's heat all day.	Paint roofs white with heat-reflecting paint. Set up shaded bus stops with drinking water.
No trees or greenery	Bad	Only 0 trees and very few parks recorded in this area.	Plant fast-growing local shade trees (Neem, Peepal) along footpaths and busy roads.
Dirty water mixing with rainwater	Medium	Many homes in slum areas don't have proper sewer pipes.	Seal and clean open drains so dirty water doesn't mix with rainwater during monsoon.

3. How Will the Money, Workers and Time Be Used?

Work To Be Done	How Many?	Money (Rs.)	Workers Needed	Time Needed	Who Is Responsible?
Cool Roof Reflective Coating in High-Density Receptors	4 building cluster	Rs. 480,000	8 workers	1 mo	PMC Building & Development Department

Summary:
• Total money being used:** Rs. 480,000 out of Rs. 1,000,000
• Money left over:** Rs. 520,000
• Workers needed:** 8 person-months
• Everything done in:** 1 months

4. What Should the Ward Officer Do? (Step by Step)

1. **Week 1-2:** The ward engineer should visit the worst waterlogging spots and get the mud and garbage cleared from major drain channels.
2. **Month 1:** Work with local community leaders to identify tin-roof buildings and get them painted with white heat-reflecting coating.
3. **Month 2:** Coordinate with PMPML to set up shaded bus shelters with clean drinking water taps at the busiest stops.
4. **Month 3:** The Garden Department plants hardy local shade trees (Neem, Peepal) along road edges with protective tree guards.

5. How Will Residents Know Things Are Getting Better?

• ■ **Streets clear up quickly:** After heavy rain, water should drain away within 30-45 minutes instead of staying for hours.
• ■ **Homes feel cooler:** Rooms under white-painted roofs should feel noticeably more comfortable during hot afternoons.
• ■ **Bus stops with shade and water:** Commuters and pedestrians can use shaded bus stops and new tree saplings are alive and growing.